

Examen final

1) $m = 8 \text{ g}$ $M = 250 \text{ g}$ $h = 1 \text{ m}$
 $d = 2 \text{ m}$

$$V = d \frac{(m+M)}{m} \sqrt{\frac{g}{2} h}$$

$$V = 2 \frac{(8+250)}{8} \sqrt{\frac{9.8}{2} \cdot 1} = 142.78$$

2) $K = \frac{1}{2} (m_1 + m_2) V_F^2$

$$\begin{aligned} m_1 V_1 + m_2 V_2 &= m_1 + V_F \\ m_1 V_1 &= m_1 + V_F \end{aligned} \quad \left\{ \begin{aligned} V_F &= \frac{m_1 V_1}{m_1 + m_2} \end{aligned} \right.$$

$$K = \frac{1}{2} (m_1 + m_2) V_F^2 = \frac{1}{2} (m_1 + m_2) \left(\frac{(m_1 V_1)^2}{(m_1 + m_2)^2} \right) V_F^2 = \frac{m_1 V_1}{m_1 + m_2}$$

$$K = \frac{1}{2} \frac{m_1^2 V_1^2}{m_1 + m_2}$$

$$\frac{1}{2} \frac{m_1^2 V_1^2}{m_1 + m_2} = (m_1 + m_2) gh$$

$$m_1^2 V_1^2 = 2 (m_1 + m_2) (m_1 + m_2) gh$$

$$m_1^2 V_1^2 = 2 (m_1 + m_2)^2 gh$$

$$V_1^2 = \frac{2 (m_1 + m_2)^2 gh}{m_1^2}$$

$$V_1 = \sqrt{\frac{2 (m_1 + m_2)^2 gh}{m_1^2}} = \frac{m_1 + m_2}{m_1} \sqrt{2 gh}$$

$$h = 5 \text{ cm} = 0.05 \text{ m}$$

$$m_1 = 5 \text{ g} = 0.005 \text{ kg}$$

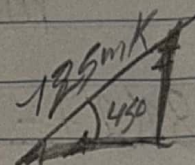
$$m_2 = 7 \text{ kg}$$

$$V_1 = \frac{(0.005 + 1)}{0.005} \sqrt{2 \cdot 9.8 \cdot 0.05} = 198.9798 \approx 198.98 \text{ m/s}$$

3)

$$m_1 = 5000 \text{ kg}, m_2 = 200 \text{ kg}$$

$$V_2 = 88.39$$



$$m_1 V_1 + m_2 V_2 = m_1 V_1 + m_2 V_2$$

$$0 + 0 = 5000(V_1) + 200(88.39)$$

$$V_x = \cos 45 (125)$$

$$V_x = 88.39 \text{ m/s}$$

$$V_1 = 3.54 \text{ m/s}$$

4)

$$-Fd = \frac{1}{2} m (V^2 - V_0^2)$$

$$m = P/g$$

$$F = \frac{-8820(35)^2}{-9.8 \cdot 2 \cdot 1000} = 557.25 \text{ N}$$

$$5) m \cdot 30 - m \cdot 20 = mV + mU$$

$$30 - 20 = 10 = V + U$$

$$30 - (-20) = -(V - U)$$

$$U = 50 - 20 = 30 \text{ cm/s}$$

$$50 - 10 = -2V$$

$$V = \frac{-40}{2} = -20 \text{ cm/s}$$

$$\begin{aligned} \textcircled{c)} \quad m_1 &= 2.7 \text{ Kg} \\ m_2 &= 2.7 \text{ Kg} \\ \mu &= 0.15 \\ \theta &= 25 \end{aligned}$$

$$m_2 g - T = m_2 a$$

$$T - m_1 g \sin \theta - \mu m_1 g \cos \theta = m_1 a$$

$$m_2 g - m_1 g \sin \theta - \mu m_1 g \cos \theta = (m_2 + m_1) a$$

$$a = \frac{g(m_2 - m_1 \sin \theta - \mu m_1 \cos \theta)}{m_2 + m_1}$$

$$a = \frac{9.81(2.7 - 2.7 \sin(25) - (0.15)(2.7)(\cos 25))}{(2.7 + 2.7)}$$

$$a = 2.16524 \approx \underline{\underline{2.17 \text{ m/s}^2}}$$